TSM 098 Arithmetic Test - Practice 1

Version 202610122020.1 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Evaluate at x = -2, y = 3. Show the substituted expression.

$3x - 4y - 2$

2. Complete the table by evaluating each side separately. State what the tested values establish; do not solve the equation.

$x(x + 2) = x^2 + 2x$

x	Left value	Right value	Equal?
2			
-2			
0			

3. Evaluate each expression.

$(a)\quad 9 - (-5)$

$(b)\quad (-6)(-4)$

$(c)\quad \frac{7}{0}$

$(d)\quad -8 - 5$

4. Evaluate the same expression at (a) x = 10, (b) x = -10, and (c) x = 0. Show substitution.

$x^2 - 3x + 2$

5. Reduce each fraction to lowest terms.

$(a)\quad \frac{14}{21}$

$(b)\quad \frac{42}{66}$

Student answer sheet

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Question 1

Value	20 pts

Question 2

x = 2	4 pts
x = -2	4 pts
x = 0	4 pts
Conclusion	8 pts

Question 3

(a)	5 pts
(b)	5 pts
(c)	5 pts
(d)	5 pts

Question 4

(a)	7 pts
(b)	7 pts
(c)	6 pts

Question 5

(a)	10 pts
(b)	10 pts

Answer key

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Question 1

Value	-20	20 pts

Question 2

x = 2	Left 8; right 8; yes	4 pts
x = -2	Left 0; right 0; yes	4 pts
x = 0	Left 0; right 0; yes	4 pts
Conclusion	All tested values work; this alone does not prove an identity.	8 pts

Question 3

(a)	14	5 pts
(b)	24	5 pts
(c)	Undefined	5 pts
(d)	-13	5 pts

Question 4

(a)	72	7 pts
(b)	132	7 pts
(c)	2	6 pts

Question 5

(a)	$\frac{2}{3}$	10 pts
(b)	$\frac{7}{11}$	10 pts